$$\dot{D} = \frac{dD}{dt} \overset{CEP}{=} \frac{d\,K_c}{dt} = \frac{d}{dt}\left(\underline{\Psi} \cdot \frac{\mu_{tw}}{\rho}\right)_{E,z}$$

$$= \frac{d\underline{\Psi}}{dt} \cdot \frac{\mu_{tr}}{\rho} = \sim$$